

Introduction to Deep Learning

Machine Learning is a branch of artificial Intelligence that focuses on developing models and algorithms, that let computers learn from some historical data without being explicitly programmed for every task.

The process of machine Learning begins with the raw data which is used for extracting useful information that helps in decision-making.

Supervised Learning adapts a system such that for a given input data it produces a target output.

The Learning data is made up of tuple (attributes, Label) where "attributes" represent the input data and "Label" represent the target output.

Supervised Learning can use both continuous and discrete types of input data.

Unsupervised Learning involves data that comprise of input vectors without any target output, such as clustering, density estimation, and visualization.

The goal of clustering is to discover groups of similar data items on the basis of measured or perceived between of similar data item.

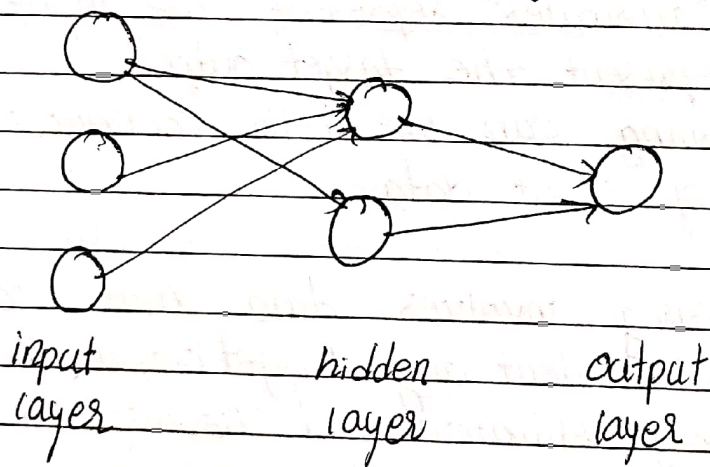
In Visualization, the data is projected down from a high dimensional space to two or three dimensions to view the similar data items.

*) Semi-Supervised learning can involve not providing with an explicit form of error at each time but only a generalized reinforcement is received giving indication how the system should change its behavior.

Shallow Learning

Shallow architectures are well understood and perform good on many common machine learning problems, that which has one input layer and one hidden layer and one output layer,

To find means to solve more complex real-world problems which shallow architecture are unable to learn models adequately.



Deep Learning

- * Deep Learning refers to the architectures which contain multiple hidden layers (deep Network) to learn different features with multiple levels of abstraction.
- * Deep learning algorithms seek to exploit the unknown structure in the input distribution in order to discover good representations, often at multiple levels, with higher level learned features defined in terms of lower level features.
- * Conventional machine learning techniques are restricted in the way they process the natural data in the raw form.
- * Constructing pattern recognition or machine learning system required considerable domain expertise and careful hand engineering to come up with a feature extractor. The transformed the raw data, into suitable internal representation or feature vector from which the learning system.
- * Deep learning allows inputting the raw data (pixels in case of image data) to the learning algorithms without first extracting featuring data.
- * Deep learning algorithms can learn the right set of features, and it does this in a much better way than extracting these features from raw data by a set of rules and algorithms.

1) deep learning involves learning these features automatically during the training process.

*) The lower layers of the model encode some basic representation of the problem, whereas higher layers build upon the lower layers to form more complex concepts.

2)

fig of Deep learning

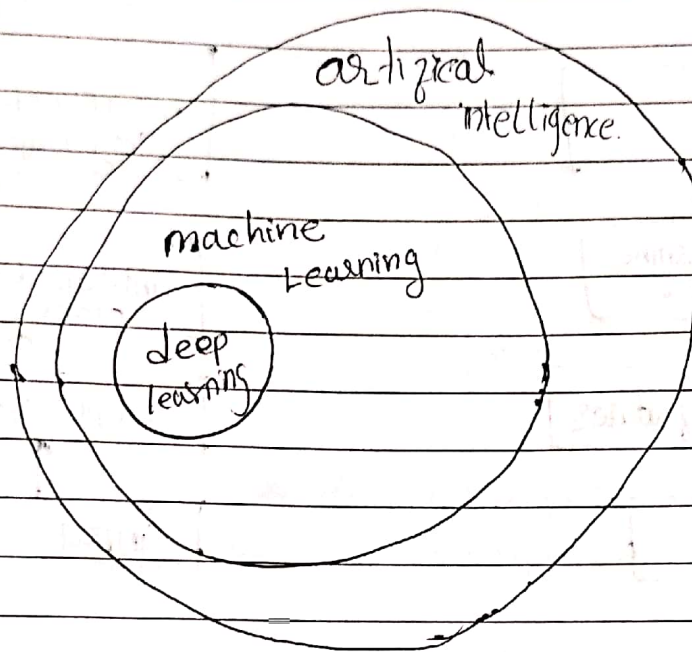
*) Given an image, the pixel intensity values are fed as inputs to the deep learning system. A hidden layer then extracts features from the input image. These hidden layers are built upon each other in a hierarchical fashion.

*) The lower level layers of the network detect only edge-like regions. These edge regions are then used to store the information of current stage and corner to store the information of next stage.

*) The number of layers used to model the data determines the depth of the model. Current deep learning used to model the learning tens or even hundreds of

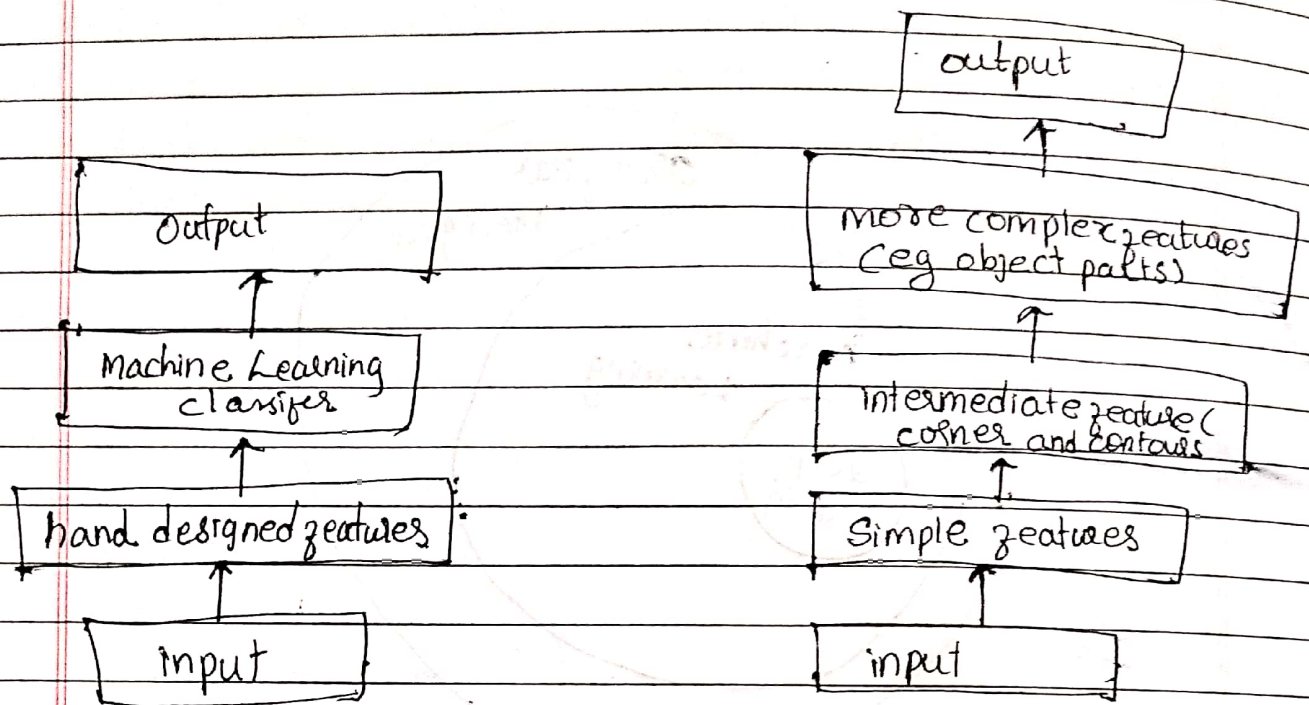
atically

layers of representation from the training data automatically.



Relationship b/w AI, ML and DL.

- *) Deep learning and machine learning are sub-fields of AI, ML, and DL.
- *) The successive layers of representations may be learned via sub-models, which are structured in the form of layers stacked on the top of each other.
- *) As deep learning network has typically more layers and parameters, it has the potential to represent more complex input.
- *) Although deep learning has been around since 1980's, it was relatively unpopular for several years as the computational infrastructure.



(a) Machine Learning

(b) Deep Learning

- * The layers in the higher level combine corner and contour to lead to more abstract "object parts" in the next layer.
- * The output layer classifies the image and obtains the output class label - the output obtained at the output layer is directly influenced by every other node available in the network.
- * In deep learning meaningful representations from the input data are learnt by putting emphasis on building complicated mapping using a series of simple learning.

- * The number of layers used to model the data determines the depth of the model.

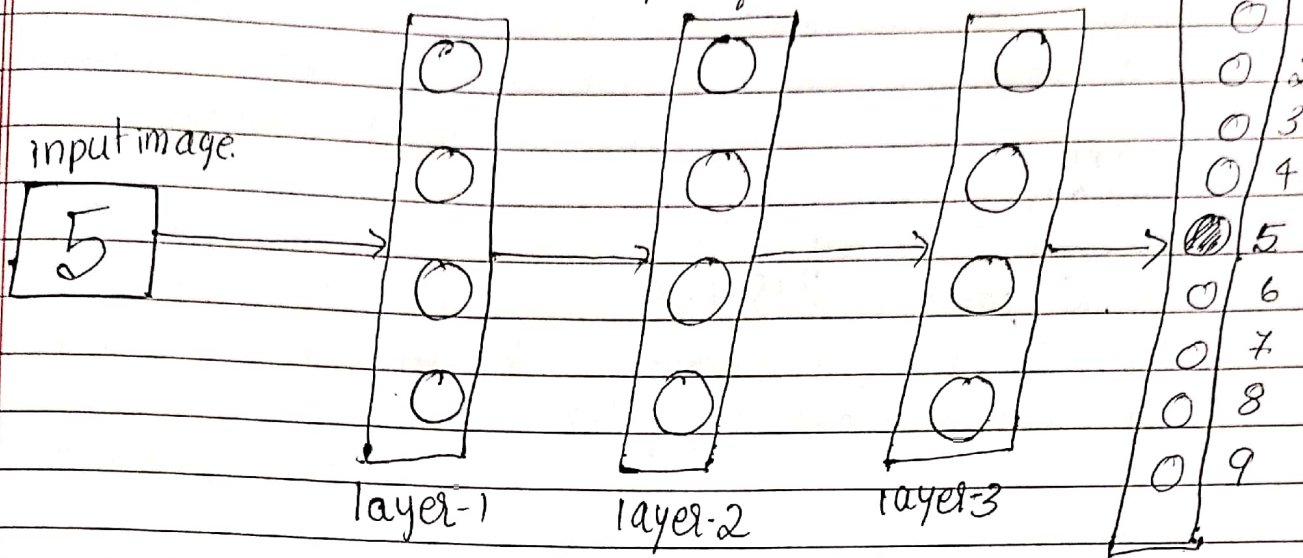


Fig: A deep learning network for digit classification.

- * Improved computational resources for processing massive amount of data and training much larger model.
- * Automatic features extraction.
- * Deep Learning models are formulated by only drawing inspiration from the understanding of biological brain.
- * The Learning mechanisms used by deep learning model are in no way comparable to the human brain, but can be described as a mathematical framework for learning representation from data.
- * The deep network transforms the digit image into representations that tend to capture a higher-level of abstraction. Each hidden layer transforms the input image into a representation that is increasingly different from the original image.

and increasingly informative about the final result.

input
image:

5 → Fig 1.4

*) Deep Network can be thought of as a multistage distillation information operation, where layers use multiple filters on the information to obtain an increasingly transformed form of information.

*) It is a fast-growing field that circumvent the problem of feature extraction which is used as a prelude by conventional machine learning approaches.

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Why to use Deep Learning.

In making it difficult to create computer learning systems that can respond to high-dimensional input and perform hard AI tasks.

- *) At the time of Big Bang introduction of deep learning, the state-of-the-art machine learning algorithms had already took decades of human effort to accumulate relevant set of features required to classify the input.
- *) Deep-learning has surpassed those conventional algorithms in accuracy as the features are learnt from the data using a general-purpose learning procedure instead of being designed by human engineers.
- *) Today, many tech giant company - Facebook, Baidu, Amazon, Microsoft and Google - have commercially deployed deep learning application.
- *) The companies have vast amount of data and deep learning works well whenever there are vast volumes of data and complex problem to solve.

Some of application of deep learning.

Near-human-level image classification.

Near-human-level speech recognition.

Near-human-level handwriting transcription.

Improved self-driving car.

Improved search results on the web.

* improved ad targeting, as used by google, Baidu and ~~Bing~~ Bing.

* Ability to answer natural language questions.

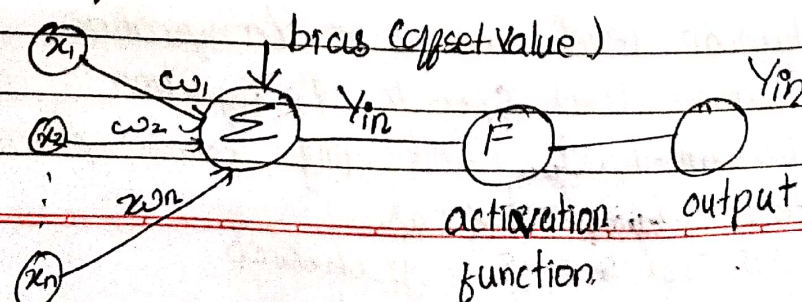
* The exceptional performance of deep models can be mainly attributed to their flexibility in representing a rich set of highly non-linear functions as well as the devised methods for efficient training of these powerful networks.

* can take advantage of vast amount of data and computation power. Deep learning has succeeded in previously unsolved problems which were quite difficult to resolve using machine learning as well as other shallow network.

How Deep learning works.

* Deep networks map input to target via a sequence of layered transformation and that these layered transformations are learned by exposure to training.

* The transformations that a layer applies to its input are determined by the layer's weights, which are basically a bunch of numbers.



- *) Learning can be defined as the process of finding the values of the weights of all layers in the network in such a manner that input examples can be correctly mapped to their associated targets.
- *) A deep network one needs to find out how far the calculated output of the network is from the desired value. This measure is obtained by using a loss function (objective function).
- *) The objective function calculates the difference between the predicted output obtained from the network and the true target value for a specific example.
- *) The objective of the training is to find the values for the weights that minimize the chosen error function.
- *) Backpropagation algorithm involves assigning random values to the weight vectors, the output obtained from the network can be far from what it should be, and accordingly the loss score may be very high.
- *) This process is repeated a number of times, until the weight values that minimize the loss function are obtained.

Deep learning challenges.

- i) Deep learning networks have brought their own set of problems and challenges which outweighed.
- ii) Training these architectures for general use was impractically slow.
- iii) With limited computational power, deep learning networks were already overtaken by other approaches kernel methods.
- iv) With the significant growth in computational power and access to large labeled datasets paved the way for its return.
- * The strategies reported in the literature that deal with the difficulties of training deep networks include developing better optimizers, using well-designed initialization strategies, using activation functions